DAY 05 • STUDENT ACTIVITIES

Checking for Alerts

Receiving detection alerts and incident reports, and checking for red-flag detections as per SOP

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats
Coverage	Element 1 Check for alerts: Performance Criteria 1.1–1.3; Range of Variables: Detection alert, Detection report, Red flag detection; Required Knowledge 1.5 Sources of detection alert / incident reports, 1.6 Log and detection management, 1.7 Security solution scan and operations procedure, 1.10 Malicious software behaviors
Duration	8 hours (1 training day)

Trainee Name	Batch / Section	Date

Activities Overview

No.	Activity	Type	Time	Assets Needed
5.1	Receive Detection Alerts per SOP	Individual (hands-on)	60 min	soc_alert_sop.txt, alert_queue.csv (lab: ~/day5)
5.2	Receive Incident Reports	Individual	45 min	incident_reports.txt, soc_alert_sop.txt (lab: ~/day5)
5.3	Red Flag Check: Ransomware or PE Infection?	Individual (hands-on)	75 min	edr_file_events.csv, soc_alert_sop.txt (lab: ~/day5)
5.4	Check for Alerts on Your Workstation	Individual (hands-on)	45 min	CTM Virtual Lab: your WKS-LAB workstation (day5\my_edr_events.csv, Documents)

CAUTION — Lab Rules

- Work only on the lab machines, virtual machines and files provided by your trainer.
- Never scan, probe or test any system outside the authorized training network.
- Treat all sample logs and data as confidential, as if they belonged to a real client.
- The ransomware and virus evidence is simulated. Never download or run real malware samples.

Activity 5.1 — Receive Detection Alerts per SOP

Type	Individual (hands-on)
Time Allotment	60 min
Objective	Receive and record the shift's detection alerts against the SOP time limits, and find the red flags.
Assets / Materials	assets/soc_alert_sop.txt; assets/alert_queue.csv (CTM Virtual Lab: ~/day5/)

Instructions

1. It is 09:30 on your shift. Read SOC-SOP-01 Section A, especially the time limits in rule A2.
2. Open alert_queue.csv. In the CTM Virtual Lab use Import-Csv day5\alert_queue.csv | Where-Object status -eq New | Format-Table, or grep ",New," on Ubuntu.
3. For every alert, work out the minutes: acknowledged time − received time; for a New alert, 09:30 − received time.
4. Compare the minutes with the limit for its severity: within the SLA, an SLA breach, or N/A (Informational).
5. Read Section C and mark the alerts that are red flags. Decide which alert you must handle first right now.

Work Space

Alert ID	Severity / Status	Minutes	Within SLA?	Red Flag? (code)
Q-5001				
Q-5002				
Q-5003				
Q-5004				
Q-5005				
Q-5006				
Q-5007				
Q-5008				
Q-5009				
Q-5010				
Q-5011				
Q-5012				
Q-5013				

Number of New alerts	

Acknowledged too late	
Handle first now, and why	
Red-flag alerts and codes	

Deliverable

Completed alert receipt log (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Recorded every alert with the fields required by the SOP	☐	☐
Correctly identified the SLA breaches, including the late acknowledgment	☐	☐
Identified the ransomware and PE infection red flags and put the Critical red flag first	☐	☐

Activity 5.2 — Receive Incident Reports

Type	Individual
Time Allotment	45 min
Objective	Record incident reports from six channels as per SOP, and recognize red flags and social engineering.
Assets / Materials	assets/incident_reports.txt; assets/soc_alert_sop.txt (CTM Virtual Lab: ~/day5/)

Instructions

1. Read SOC-SOP-01 Section B (reports from people).
2. Open incident_reports.txt. In the lab: grep -E "^(=== REPORT|CHANNEL)" ~/day5/incident_reports.txt, then grep -A9 "REPORT IR-01" (change the ID).
3. For each report, record the channel, reporter and callback number, and the affected computer or account.
4. Classify it: security incident with a red flag, security incident, social engineering attempt, or not a security matter.
5. Choose the next step the SOP requires, and find the two reports that match alerts from Activity 5.1.

Work Space

Report	Channel	Reporter / Callback	Type	Next Step per SOP
IR-01				
IR-02				
IR-03				
IR-04				
IR-05				
IR-06				

Reports that match Activity 5.1 alerts	
First-response advice to the IR-01 caller	

Deliverable

Completed intake records (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Recorded the channel, reporter, callback and affected asset for all six reports	☐	☐
Escalated the two red flags immediately	☐	☐

Criteria	Yes	No
Refused the OTP request and treated it as social engineering	☐	☐
Redirected the non-security report to the IT Helpdesk	☐	☐

Activity 5.3 — Red Flag Check: Ransomware or PE Infection?

Type	Individual (hands-on)
Time Allotment	75 min
Objective	Check endpoint file events for ransomware and PE infection red flags, and rule out approved look-alikes.
Assets / Materials	assets/edr_file_events.csv; assets/soc_alert_sop.txt (CTM Virtual Lab: ~/day5/)

Instructions

1. Count the actions per host: cut -d, -f2,5 ~/day5/edr_file_events.csv | sort | uniq -c
2. Read the Rename and Create rows: what changed, which process did it, and is the process signed?
3. Look for ransom notes and backup deletion: grep -E "README|vssadmin" ~/day5/edr_file_events.csv
4. In PowerShell, group the events by signer and list the antivirus detections (Where-Object detection -like "Virus*").
5. Decide for each host: ransomware red flag, PE infection red flag, or no red flag (approved software), and write the SOP codes.

Work Space

Host	Verdict	Red-Flag Codes
FIN-PC-03		
ENG-PC-11		
QA-PC-05		
FS-02		
HR-LT-07		

FIN-PC-03: files renamed	
New extension and ransom note	
First rename: time and process	
ENG-PC-11: installers modified	
Where the PE infection started	
Why FS-02 and HR-LT-07 are not red flags	

Deliverable

Completed red flag check (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Identified the ransomware host with correct evidence (count, extension, note, first event)	☐	☐
Identified the PE infection and where it started	☐	☐
Ruled out the approved backup agent and Windows Update by process and signer	☐	☐

Activity 5.4 — Check for Alerts on Your Workstation

Type	Individual (hands-on)
Time Allotment	45 min
Objective	Apply the SOP red flag check to your own lab workstation and take the action the SOP requires.
Assets / Materials	CTM Virtual Lab: your WKS-LAB workstation (C:\Users\<you>\day5\my_edr_events.csv, Documents folder)

Instructions

1. On your WKS-LAB PowerShell tab: Import-Csv day5\my_edr_events.csv | Group-Object signer
2. List every event: Import-Csv day5\my_edr_events.csv | Format-Table time, process, action, path, new_path
3. Look at your Documents folder: Get-ChildItem Documents
4. Decide whether there is a red flag, count the affected files, and record the first event time and the process.
5. Choose the SOP action. Every seat has different evidence, so your answers will not match your classmates'.

Work Space

Red flag found? (type and codes)	
Files renamed or modified	
First event time	
Process	
Extension and note / detection / approved software	
Action per SOP	

Deliverable

Completed workstation check (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Correct verdict with evidence from own workstation	☐	☐
Chose the SOP action (escalate a red flag; record approved activity)	☐	☐

Self-Check

A. Multiple Choice

Encircle the letter of the best answer.

1. A Critical alert was received at 09:10. Under SOC-SOP-01 it must be acknowledged by:
    a) 09:25
    b) 09:40
    c) 11:10
    d) The end of the shift
2. Which of these is a ransomware red flag?
    a) Windows Update replaces DLLs in System32
    b) Many files renamed to the same new extension
    c) A backup agent creates .scbak files
    d) A user downloads a PDF
3. A "PE infection" is malware that:
    a) Steals saved browser passwords
    b) Inserts its code into .exe and .dll files
    c) Floods a website with traffic
    d) Only shows advertisements
4. An SMS from "IT" asks you to send the OTP that will arrive on the SOC phone. You should:
    a) Send it quickly because it is urgent
    b) Delete it and say nothing
    c) Not comply, call back on the directory number, and report it
    d) Forward it to all staff
5. According to the Competency Standards, a detection report may come by:
    a) Phone, walk-in, email, SMS, chat or video conference
    b) SIEM only
    c) Email only
    d) Firewall and WAF only

B. True or False

Write TRUE if the statement is correct, FALSE if it is not.

No.	Statement	Answer
1	A Level I analyst may delete an alert that looks unimportant.	
2	If ransomware is suspected, the user should shut down the PC at once.	
3	A printer jam reported by chat is recorded and redirected to the IT Helpdesk.	
4	Approved software signed by its vendor doing its normal job is not a red flag.	

C. Short Answer

1. List the four ransomware red flags in SOC-SOP-01.

2. Why do we call back on the directory number instead of replying to the number that contacted us?

Reflection

What was the most important thing you learned today, and how will you apply it in a Security Operations Center?

Trainer Sign-off

Overall Result	Trainer Remarks	Trainer Signature / Date
☐ Competent		
☐ Not Yet Competent		